

Does the Handover Margin Earn Its Place?

A Factorial Comparison of Temporal Decision Policies for Context-Based Continuous Smartphone Authentication

The margin did not earn its place: a short dwell captured most of the measurable stabilisation, at lower cost.

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Scores are not states

A classifier emits a score every frame. A device must hold a persistent state: authenticated, or locked.

Scores that sit near the threshold make the state flicker even when the user has not changed.

A high AUC cannot detect this. AUC ranks frames; it says nothing about the behaviour of the state over time.

Instantaneous threshold



8 state changes - same user, same scores

With a persistence requirement



1 state change - same user, same scores

Filled = authenticated. Illustrative schematic, not measured data; real traces appear on the next slides.

The gap and the question

Stabilisation is usually asserted, not decomposed. Zeeshan et al. (2025) use EWMA "to stabilize decisions and avoid flickering" - without isolating its effect.

Mondal and Bours (2015) do compare decision layers, but sweep four factors together and report action-domain measures, not a matched-operating-point contrast.

Sugrim et al. (2019) show why equal-error-rate comparison is uninformative when a system has a fixed false-accept requirement.

Research question

At a calibration-defined decision-layer FAR target, does adding a handover-inspired margin to temporal dwell reduce excess authentication-state transitions without unacceptable false rejection or detection failure?

What transfers from handover

Cellular handover

Switch when a neighbour cell exceeds the serving cell by a hysteresis margin, and holds that lead for a time-to-trigger.

Continuous authentication

Change state when the score crosses a dual threshold, and stays across k consecutive frames.

Shared

Noisy sequential evidence - costly state reversals - a responsiveness against stability trade-off

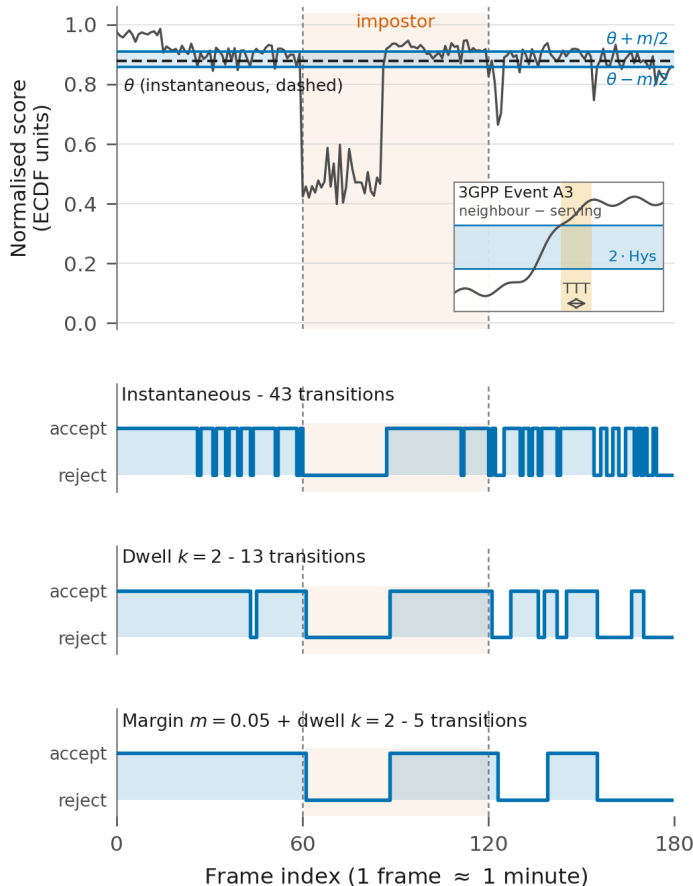
Not shared

Handover arbitrates competing radio links. Authentication weighs identity hypotheses, where false accept and false reject are asymmetric security outcomes.

Terminology: in the analogy the hysteresis term is the margin; time-to-trigger is the dwell. The composed policy is called margin + dwell.



The evaluation contract



ExtraSensory, 31-participant regular-cadence cohort. Chronological enrolment, calibration and test partitions; final-test impostors unseen during fitting and calibration.

One frozen per-user gradient-boosting score stream. Every policy is replayed offline on identical scores, so any difference is the decision rule, not the classifier.

Each test sequence: 60 genuine, 60 unseen-impostor, 60 recovery frames.

**A correct trace needs exactly two state changes.
Anything beyond that is an excess transition.**

Figure 5. Controlled identity-transition sequence, real observations, composed to a fixed structure.

Design and preregistered test

margin $m \in \{0, 0.05, 0.10, 0.20\}$ $\times$ dwell $k \in \{1, 2, 3, 5, 10\}$

Instantaneous

$m = 0, k = 1$

Dwell only

$m = 0, k > 1$

Margin only

$m > 0, k = 1$

Margin + dwell

$m > 0, k > 1$

Parameters chosen on calibration data only, at a calibration-defined decision-layer FAR target of 0.05 (feasible band 0.04-0.05).

The participant is the unit of inference; paired tests, Holm-corrected.

Margin + dwell earns its place only if

1. fewer excess transitions after Holm correction
2. FRR non-inferior: CI upper bound below +0.02
3. detection failure non-inferior

against BOTH dwell only and margin only.

Rules, grids, outcomes and the verdict wording were frozen before the confirmatory score dumps existed.



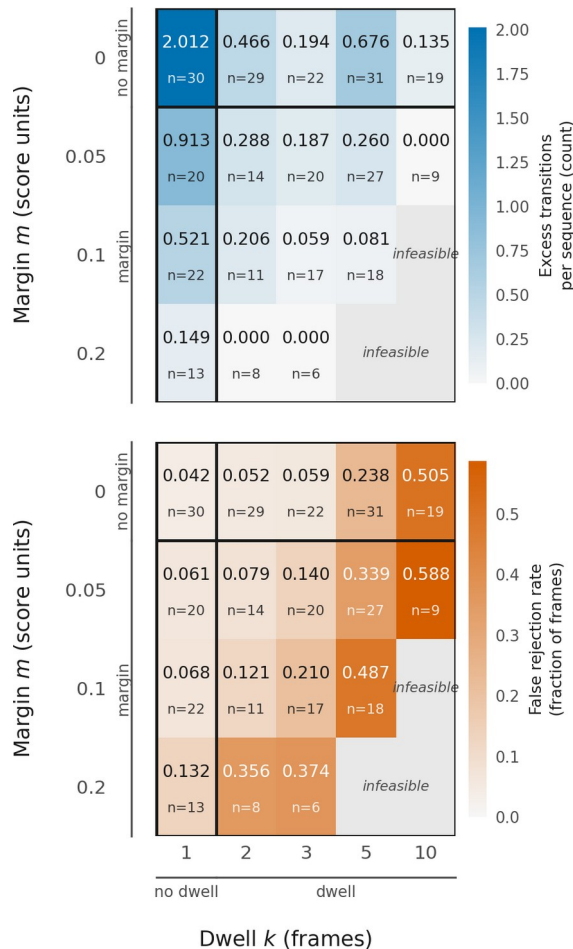
Primary result

	vs dwell only	vs margin only
Participants paired	30	24
Excess transitions, mean paired difference	-0.13	-0.20
Median paired difference	0	0
Holm-adjusted p	0.69	0.69
FRR difference [95% CI]	+0.074 [0.032, 0.119]	+0.030 [-0.008, 0.069]
Detection-failure difference	0.000	-0.007

Margin plus dwell offers no measurable advantage over its components on this benchmark, and is FRR-inferior to dwell alone.

Against dwell, 20 of 30 participants tied exactly. The +0.074 FRR penalty breaches the preregistered +0.02 bound.

Reading the surface



**Stability improves down and to the right.
Lockout worsens along the same direction.**

Instantaneous (0, 1): 2.012 excess transitions, FRR 0.042

Dwell (0, 2): 0.466 excess transitions, FRR 0.052

Margin + dwell (0.20, 2): 0.000 excess, FRR 0.356

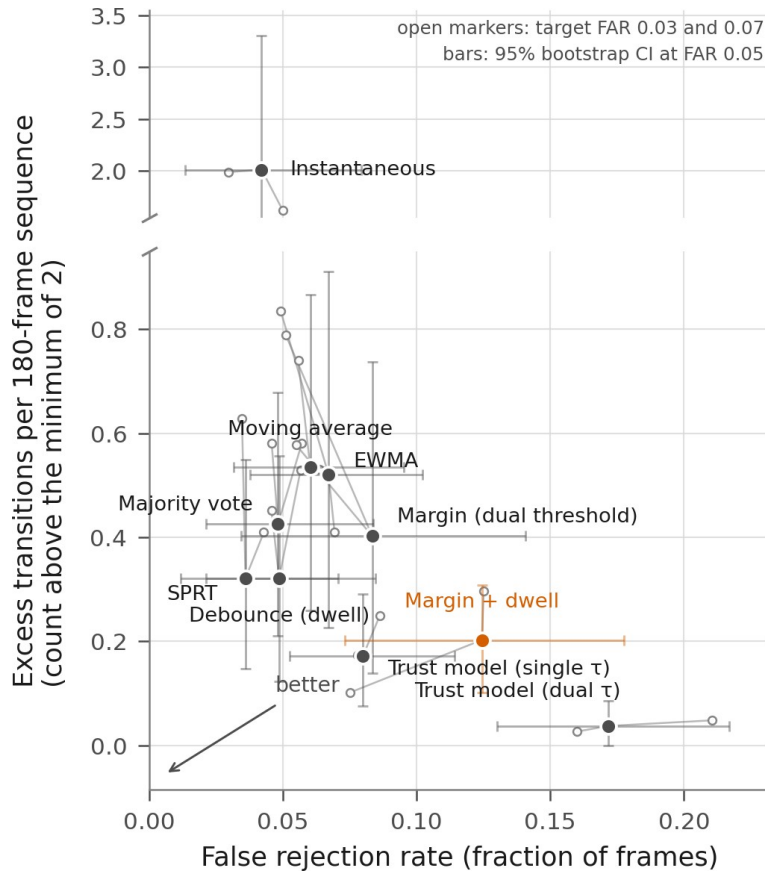
Zero flips is not success. The zero-transition cells are supported by as few as 8 participants and reject the genuine user more than a third of the time.

**Cells are NOT paired: feasible participant sets differ from 6 to 31.
Read the broad shape only.**

Figure 1. Response surface at target FAR 0.05.



No mechanism escapes it



Nine temporal mechanisms, all calibrated to the same FAR target. Lower left is better.

Every mechanism beats instantaneous on stability. None dominates on both axes.

The most stable are the most locked out: margin + dwell 0.20 excess at FRR 0.124; trust model 0.04 at FRR 0.172.

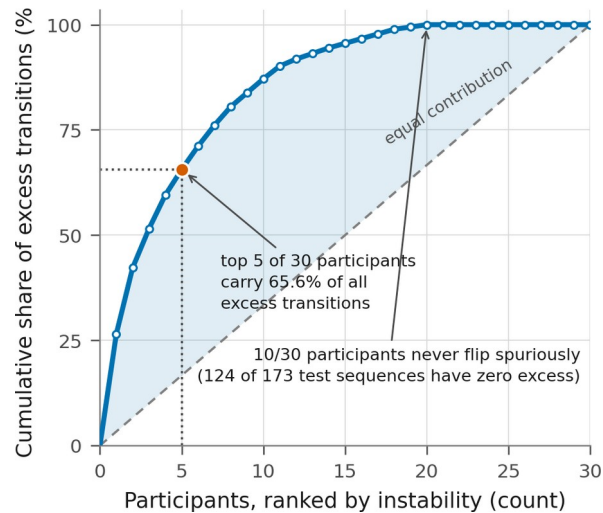
Debounce - a short dwell - is the simplest defensible first intervention.

SPRT is exploratory: it had the largest parameter grid, so the greatest selection opportunity.

Figure 2. Stability-lockout plane; small markers are FAR 0.03 and 0.07.

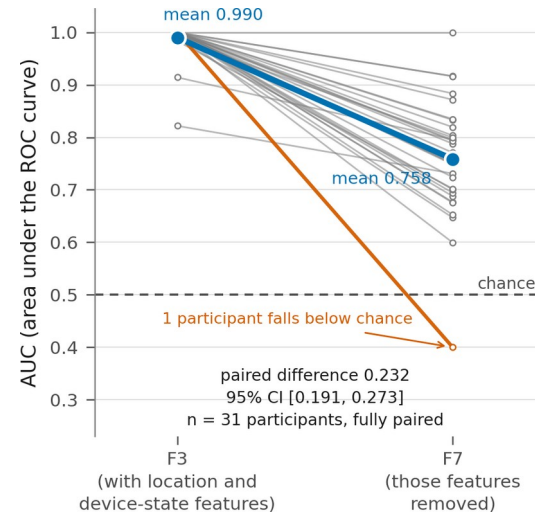


Threats to validity



Instability is rare and concentrated

71.7% of instantaneous test sequences never flip. Five of 30 participants carry 65.6% of it. There is little for a margin to improve, which limits power.



The score is not clean identity evidence

Removing location and device-state features drops mean AUC from 0.990 to 0.758; paired difference 0.232 [0.191, 0.273].

ExtraSensory is a context-recognition dataset, not an authentication dataset. The sequences are real observations, composed - not observed takeovers.

Contribution and answer

Evidence

A preregistered null: no stability gain from the margin ($p = 0.69$ against both components), and an FRR penalty of $+0.074$ against dwell that breaches the $+0.02$ bound.

Interpretation

Persistence is what transfers from handover. The margin does not add measurable value here - it converts instability into lockout.

Recommendation

Evaluate the decision layer separately from the classifier, and always report excess transitions with FAR, FRR and recovery together.

The margin did not earn its place on this benchmark.

